Two-dimensional powder diffraction in layered van der Waals films: quantitative X-ray area-detector modeling

1 Introduction

For layered thin films, the central structural question is often not only which phase is present, but how the crystallites are oriented relative to the substrate. In layered van der Waals solids and related two-dimensional materials, this orientational state can directly control the properties that motivate the films in the first place.[1–3] In layered halides such as PbI_2, transport and device response can depend strongly on whether the layers lie parallel to the substrate.[4] In topological bismuth chalcogenides, strong c-axis alignment is needed to preserve the desired electronic response.[5] In transition-metal dichalcogenides, catalytic activity can increase when platelet edges are exposed by the film orientation.[6] Comparable orientational states also occur across a wider set of layered thin-film families, including telluride chalcogenides (e.g., Bi_2Te_3, Sb_2Te_3), dichalcogenides (e.g., MoS_2, NbS_2), layered halides and oxyhalides (e.g., BiI_3, BiOI), and laminated 2D-flake films such as $Ti_3C_2T_x$ MXenes and graphene-oxide papers.[7–14]

Wide-angle x-ray scattering (WAXS) with an area detector is attractive for this problem because a single exposure probes a large reciprocal-space volume rich with information of both structure and orientation.[15–19] The difficulty is that many layered films do not fall cleanly into either of the standard diffraction limits. They are neither single crystals, where reciprocal space consists of discrete points, nor fully isotropic three-dimensional (3D) powders, where orientational averaging produces spherical shells.[20] Instead, a common thin-film orientational state is the two-dimensional (2D) oriented powder, in which plate-like crystallites share an approximately aligned layer normal while remaining azimuthally randomized in-plane.[3, 20, 21] This intermediate orientational state produces detector patterns that inherit features of both limiting cases while necessitating nontraditional analysis techniques.[20]

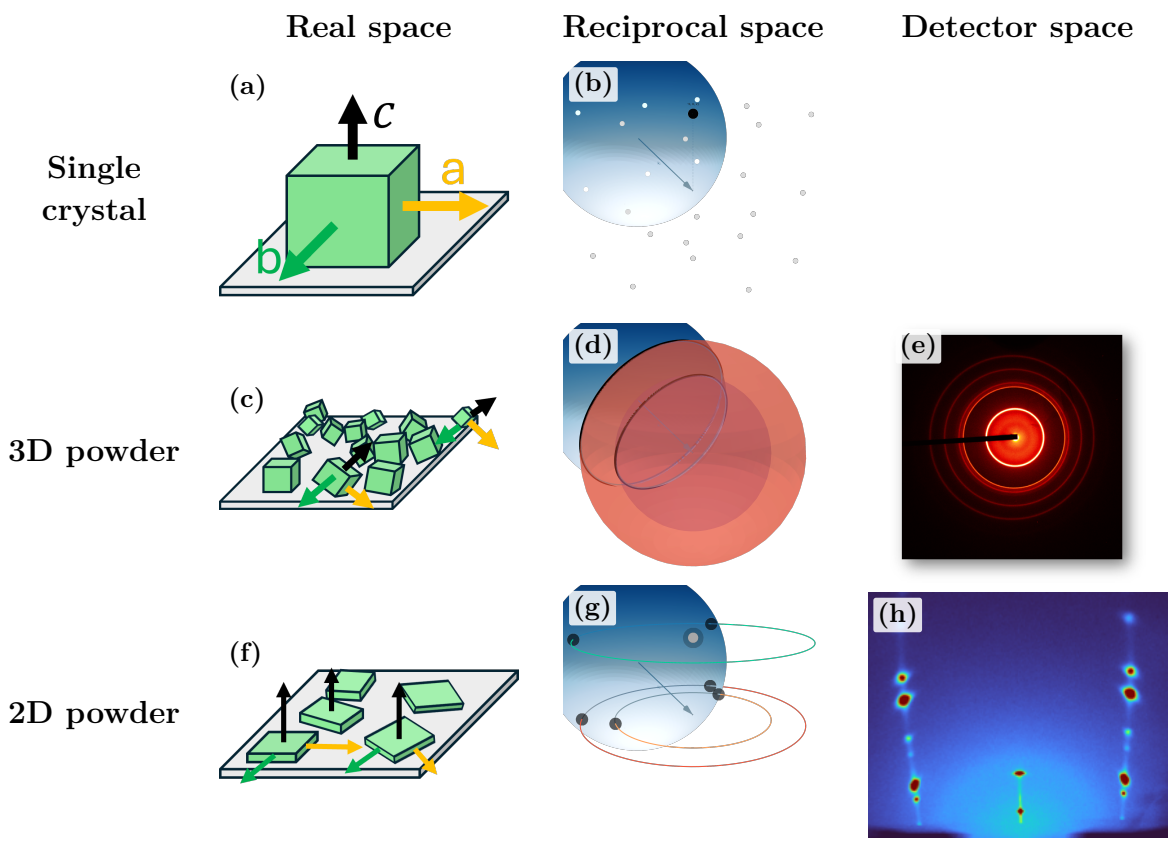

Figure 1: Diffraction across three orientational limits: single crystal, 3D powder, and 2D powder. Columns show the corresponding real-space, reciprocal-space, and detector-space views, while rows group the orientational limits. Panels (a,b) show a single crystal in real space and reciprocal space. Panels (c–e) show a 3D powder, with randomly oriented crystallites, spherical reciprocal-space shells, and Debye–Scherrer rings on the detector. Panels (f–h) show a 2D powder, with aligned surface normals and in-plane orientational disorder, reciprocal-space rings with specular Bragg features, and the corresponding detector pattern with symmetry about the central axis.

Figure 1 summarizes the real-space, reciprocal-space, and detector-space consequences of these three orientational limits. In the Ewald construction, elastic scattering requires that reciprocal-space intensity lie on the Ewald sphere of radius $|\mathbf{k}|$.[22] For a single crystal, reciprocal space consists of isolated Bragg points (Fig. 1b), so at fixed incidence angle the Ewald sphere typically intersects only a small subset of reflections and the detector records sparse spots.[22] For a 3D powder, random crystallite orientations smear the intensity of each Bragg point over a spherical shell, and the Ewald-sphere intersection produces Debye–Scherrer rings on the detector (Fig. 1d, Fig. 1e).[22] For a 2D powder, azimuthal disorder about the common c axis smears each off-specular Bragg point into an in-plane ring about Q_z, while specular $(00L)$ reflections remain localized near the Q_z axis (Fig. 1g).[15, 20] The corresponding illustration in Fig. 1f schematically shows 2D platelets with aligned growth, and a representative 2D-powder detector pattern is shown in Fig. 1h. When the Ewald sphere intersects this reciprocal-space distribution, the detector records hybrid signatures: pairs of symmetric spots from the off-specular ring intersections and single-crystal-like features near

the specular stripe from $(00L)$ intensity.[15, 17, 23] Mosaicity then redistributes intensity, broadening the ring intersections into arcs and spreading the $(00L)$ features into cap-like intensity near the specular region.[15, 20]

This hybrid detector-space information is what makes the 2D-powder problem both rich and difficult. Many standard WAXS workflows reduce the area-detector image to 1D profiles by azimuthal integration, often with masking near the direct beam and specular region.[17, 19, 24–26] Those steps are well suited to nearly isotropic powders, but for 2D powders they deweight or remove the arcs and specular caps that are most sensitive to out-of-plane correlations, stacking, and mosaic tails.[15, 17] Stacking disorder is also commonly inferred from 1D interference models and refinements,[27–29] which do not directly exploit the full two-dimensional anisotropy of oriented-film scattering. More general intermediate orientation distributions can be treated with orientation-distribution-function (ODF) refinements or preferred-orientation corrections.[30–35] Those approaches are more flexible, but they are also more general than is needed for the narrowly constrained 2D-powder case considered here.[34, 35] Recent radial-profile approaches have made thin-film GIXD substantially more quantitative with extracted one-dimensional profiles but peak overlap and background separation can remain limiting in strongly anisotropic patterns, especially at small incidence angles.[36] For the layered films considered here, that level of generality is often unnecessary. The morphology and the measured patterns already motivate an axial symmetry about the film normal together with a low-dimensional out-of-plane mosaicity kernel.[3, 20, 21] We therefore assume this physically motivated 2D-powder orientation distribution a priori and refine its parameters by forward-modeling the full 2D detector image, so that the arc and specular (Bragg-rod) line shapes in Fig. 1 act as direct constraints while reducing computational cost and improving fit precision.[15, 17]

Here we present a quantitative forward model for WAXS from two-dimensional powders that predicts full area-detector images and then refines stage-specific detector-derived observables against measured data. Prior work has established the main ingredients needed for such a framework, including grazing-incidence scattering formalisms,[37, 38] reciprocal-space mapping and indexing for oriented thin films,[15, 17, 19, 23] and forward simulation of diffraction from textured films.[39] Complementary atomistic tools have also been developed for constructing commensurate multilayer 2D heterostructure supercells and visualizing their reciprocal-space diffraction signatures, including Nookiin.[40] In contrast, the present work takes the crystallographic model as input and carries the calculation in detector space through the coupled effects of detector geometry, grazing-incidence propagation, mosaicity, resolution, and ordered structure-factor weighting. Geometry is constrained by detector-space peak positions, mosaicity by local profile shapes and specular-tail intensity, and ordered structure by selected Bragg-ROI intensities and Q_z profile overlays. The main-text validation is deliberately centered on the ordered 2D-powder limit using Bi_2Se_3 and Bi_2Te_3 films.[5, 7] Diffuse PbI_2 and stacking-disorder refinements are treated as a later extension of the same detector-space framework rather than as the primary proof of the method. By embedding geometric and sample effects within a single forward model, we avoid separat-

ing geometric corrections from structural fitting and instead refine geometry, mosaicity, and ordered intensity within a common detector-space comparison.

2 Diffraction Model

The forward model predicts the detector image produced by an indexed set of reciprocal-lattice vectors. For each reflection, the calculation determines the allowed exit rays, maps them to detector pixels, weights them by the structure factor, beam phase distribution, transmission, and redistributes their intensity according to the mosaic distribution. The same detector-space prediction is then compared with the Q_z projection profile's measured peak positions, local line shapes, and relative Bragg intensities.

2.1 Diffraction Geometry

We first consider a single incident ray striking a perfectly aligned crystallite, with no beam divergence, no refraction, and no absorption. Elastic scattering requires

$$|\mathbf{k}_i| = |\mathbf{k}_f| = \frac{2\pi}{\lambda}, \qquad \mathbf{Q} = \mathbf{k}_f - \mathbf{k}_i.$$

Diffraction occurs when the momentum transfer matches a reciprocal-lattice vector, $\mathbf{Q} = \mathbf{G}$. In the Ewald construction, this means that the reciprocal-space point for that reflection lies on the Ewald sphere of radius $|\mathbf{k}_i|$, and the simplest diffracted ray is obtained from

$$\mathbf{k}_f = \mathbf{k}_i + \mathbf{G}.$$

For a fixed $|\mathbf{G}|$, the set of possible reflection orientations forms a Bragg sphere of radius $|\mathbf{G}|$ about the reciprocal-space origin. The allowed exit vectors are found by intersecting that Bragg sphere with the Ewald sphere and then propagating the resulting exit ray to the detector.

For each trial incident ray, the model computes the sample intersection, applies the incidence geometry and optical corrections, finds the reciprocal-space intersections that satisfy elastic scattering, and propagates the resulting exit rays to the detector. Rays are Monte Carlo sampled from the probability distribution formed by the shape, bandwidth, and divergence of the beam. A second Monte Carlo sampling occurs for each ray to choose an intersection point. Footprint, detector tilt, refraction, and absorption enter as experimental resolution and intensity terms. The coordinate conventions for the lab frame, detector plane, sample rotation, and illuminated footprint are summarized in Figs. 2 and 3.

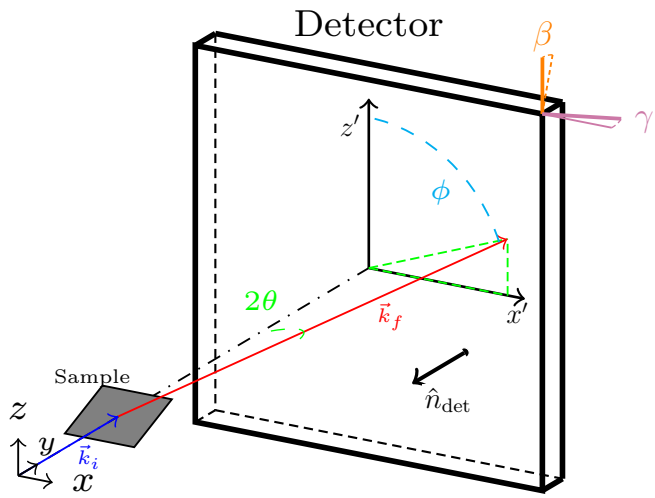

Figure 2: Overall lab-frame geometry. The incident wavevector $\mathbf{k}_i$ strikes the sample, and a representative exit wavevector $\mathbf{k}_f$ reaches the detector at scattering angle 2θ and detector-plane azimuth ϕ. The detector at distance D is tilted by β and γ.

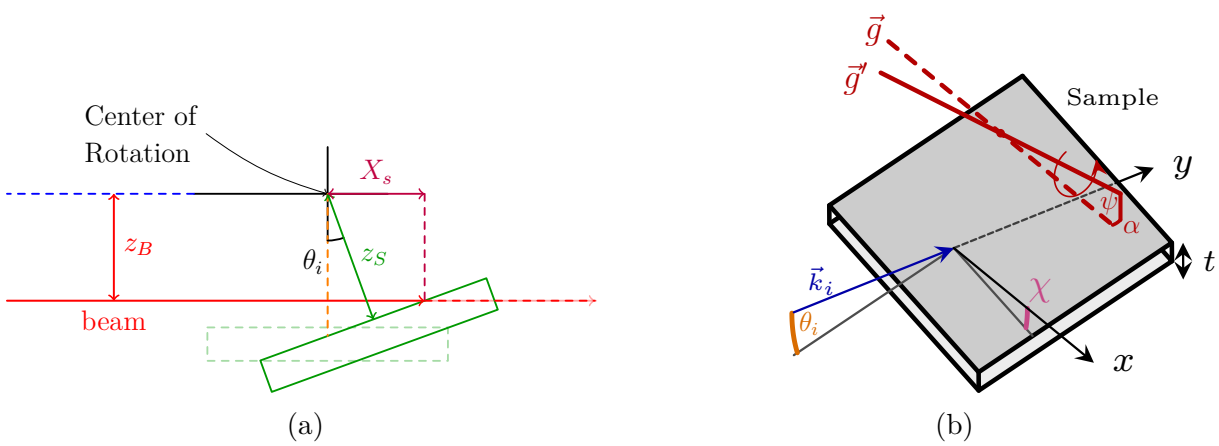

Figure 3: Two-stage sample geometry. Left: rotation about the goniometer center with sample offset z_S and beam offset z_B. θ_i is the incidence angle. Right: full alignment adds tilt δ and footprint length X_s within an illuminated window (W, H) on a film of thickness t. $\vec{g}$ is the goniometer arm (normal to $\hat{y}$), and $\vec{g}'$ is the misaligned arm obtained by rotations α and ψ.

2.2 Structure Factor

We enumerate integer in-plane Miller paire HK, construct the reciprocal-space rod along G_z for each pair, and evaluate its detector intersections. We define HK as the scalar in-plane metric sum

$$\mathrm{HK}(H, K) = \begin{bmatrix} H & K \end{bmatrix} \mathbf{M}_{\parallel} \begin{bmatrix} H \\ K \end{bmatrix} \propto Q_r^2,$$

where $\mathbf{M}_{\parallel}$ is the in-plane reciprocal-lattice metric, with any lattice-dependent normalization absorbed into the definition of HK. Thus, HK is a number labeling an in-plane reciprocal-space rod family. In powder calculations, all integer (H, K) rods with the same HK, and therefore the same Q_r, are evaluated individually and then summed. Multiplicity and orientation effects therefore emerge within the forward model.

For each reflection with reciprocal-lattice vector

$$\mathbf{G} = h\mathbf{a}^* + k\mathbf{b}^* + l\mathbf{c}^*,$$

we compute the structure factor as

$$F_{\mathbf{G}} = \sum_a o_a \, f_a(Q) \, \exp[2\pi i \, \mathbf{G} \cdot \mathbf{r}_a] \, \exp\Big[-2\pi^2 \, \mathbf{G} \cdot \mathbf{U}_a \cdot \mathbf{G}\Big] \,, \tag{1}$$

where the sum runs over atoms a in the unit cell. Here o_a, $f_a(Q)$, $\mathbf{r}_a$, and $\mathbf{U}_a$ are the occupancy, atomic form factor, fractional position, and anisotropic displacement tensor for atom a. Symmetry constraints are retained during refinement, so only the independent crystallographic parameters are varied. The detector weight for each allowed ray is multiplied by $|F_{\mathbf{G}}|^2$ before mosaic and experimental broadening are applied.

2.3 Mosaicity

Mosaicity can be understood by comparing the orientational limits introduced in Fig. 1. For a perfect single crystal, a reflection carries its full weight at a single Bragg spot. For a 3D powder, random crystallite orientations spread that weight isotropically over the Bragg sphere. A 2D powder lies between these limits. In-plane rotations are effectively randomized about the film normal, while the layer normals retain a finite out-of-plane misorientation distribution. Mosaicity is therefore treated as a redistribution of the ideal per-reflection intensity over the allowed HK family.

For a reflection, we work in the (G_r, G_z) cross-section of reciprocal space, where G_r represents the rotaions in the basal plane. Random in-plane rotations generate a ring of equivalent reciprocal-space intersections about G_z. Out-of-plane misorientation then gives a nonuniform line density around that ring, written as $I_0\omega(\Delta\theta)$, where I_0 is the reflection intensity before mosaic redistribution and $\Delta\theta$ is the crystallite tilt relative to the mean layer normal.

We will later find that a gaussian mosaic is not sufficient to reproduce the measurements. For this reason, we describe the out-of-plane mosaicity by a normalized Gaussian-plus-Lorentzian angular distribution, evaluated in wrapped form on $(-\pi, \pi]$ to conserve probability over one full angular period and avoid a discontinuity at $\Delta\theta = \pm\pi$. The Gaussian and Lorentzian widths are independent. The explicit wrapped Gaussian, wrapped Lorentzian, and surface-density forms are deferred to the supporting information.

Figure 4 summarizes how this reciprocal-space broadening appears in the $\{G_r, G_z\}$ plane

and how it maps onto characteristic detector-space features for in-plane and specular conditions.

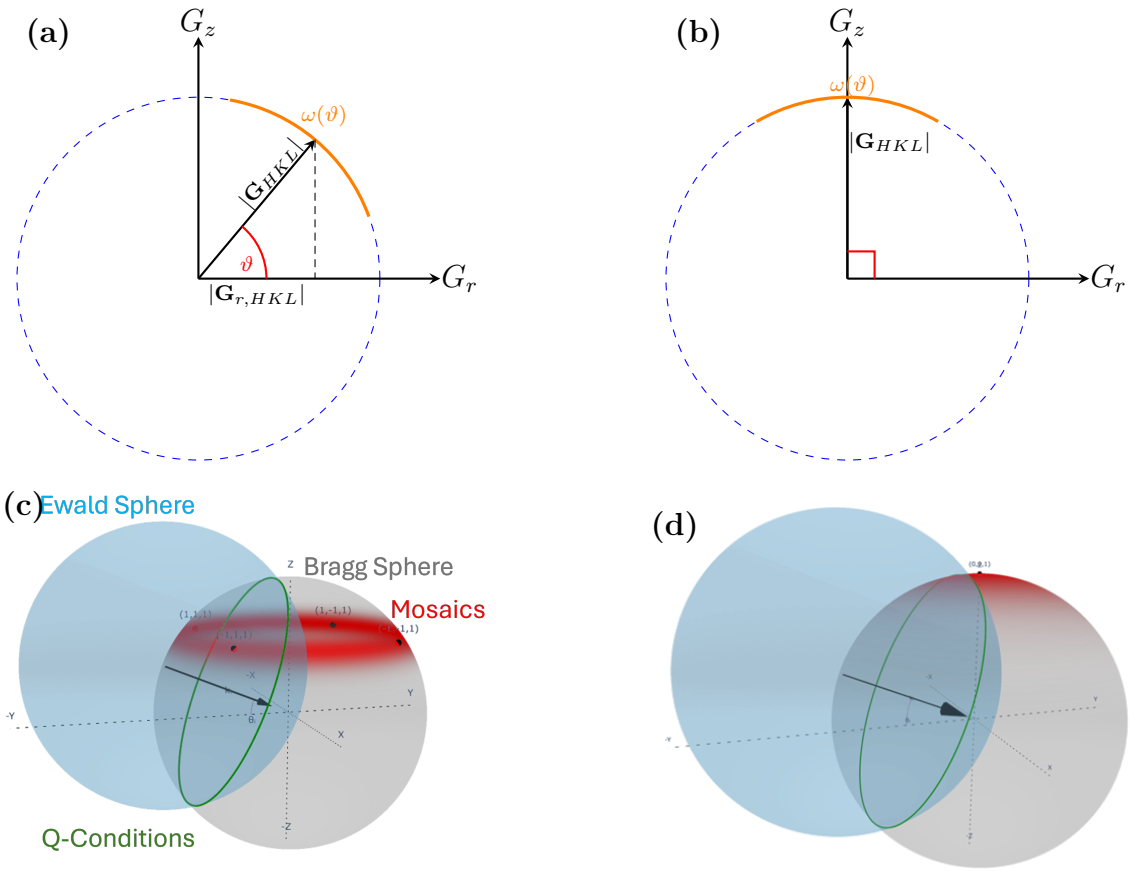

Figure 4: Mosaic broadening viewed in the $\{G_r, G_z\}$ plane (a,b) and its detector-space manifestation (c,d). Intensity is redistributed along the Bragg circle with line density $I_0\omega(\Delta\theta)$. The in-plane condition with $G_r \neq 0$ (a,c) yields two off-specular arcs, while a specular reflection with $G_r = 0$ (b,d) concentrates intensity near the polar cap. The long tail of $\omega(\Delta\theta)$ is what gives finite intensity to $HK = 0$ specular-family peaks that are not directly excited by the nominal incidence angle.

After random in-plane rotation, the one-dimensional tilt density becomes an axially symmetric surface density on the Bragg sphere. This density supplies the ray weights used to distribute I_0 over mosaic-broadened exit directions.

3 Refinement workflow

We use a staged refinement because geometry, mosaicity, and crystallographic intensity are strongly correlated in a fully global fit. Each stage uses the detector-space observable most sensitive to one parameter class before the next class is introduced. Table 1 summarizes the ordered-film workflow used for Bi_2Se_3 and Bi_2Te_3.

Table 1: Staged refinement workflow used for the ordered Bi_2Te_3 and Bi_2Se_3 films. All sample images are dark-subtracted before the listed observables are formed. Diffuse-scattering and stacking-disorder refinements are deferred from the present main-text draft.

Step	Stage	Data	Parameters refined	Primary information
1	Beam characterization	Direct-beam images at several detector distances	Beam widths w_x, w_z and divergences $\Delta\theta_x$, $\Delta\theta_z$	Incident phase space
2	Detector geometry calibration	Powder calibrant image	Detector distance D, detector tilts (β, γ), and beam center (x_0, y_0)	Detector mapping
3	Sample alignment and goniometer misalignment	Indexed reflection centers from sample images	Sample geometry $(\theta_i, \delta, z_B, z_S)$, in-plane rotation χ, goniometer misalignment (α, ψ)	Detector-space peak positions
4	Mosaicity/profile refinement	Selected local detector ROIs, specular-tail checks, and local profile shapes	Gaussian-plus-Lorentzian mosaic parameters	Center-aligned profile shapes, off-specular arcs, and specular tails
5	Ordered-film validation	Selected detector-space Bragg ROIs and Q_z projections from fixed Q_r/HK trajectories	Image scale factors and ordered structure-factor parameters	Relative Bragg intensities and measured/calculated line-shape agreement

The instrumental and alignment stages fix where an ideal reflection should appear on the detector. Direct-beam and calibrant data define the beam phase space and detector mapping, while indexed sample reflections determine the incidence condition, sample offsets, and goniometer misalignment.

The remaining stages refine what the peaks look like and how intense they are. Local detector profiles constrain the shared Gaussian-plus-Lorentzian mosaic distribution. Bragg ROI integrals constrain the ordered structure-factor parameters, and the final validation projects both measured and calculated images along the same indexed Q_r/HK trajectories for comparison versus Q_z. The ordered-film figure is therefore not a separate illustration step; it is the closure test for the workflow. Peak positions test the calibrated geometry and sample alignment, widths and tails test the two-component mosaicity and resolution, the off-condition $HK = 0$ specular peaks test the Lorentzian mosaic tail, and relative intensities test the ordered structure-factor weighting.

Implementation details required for efficiency, including 2θ–ϕ remapping, caked fields, lookup tables, and Monte Carlo sampling, are described in the supporting information.

4 Results: ordered Bi_2Se_3 and Bi_2Te_3 films

Ordered Bi_2Se_3 and Bi_2Te_3 films validate the model by showing that the same detector-space forward model quantitatively reproduces their measured line shapes and intensities. Together, they test the 2D-powder orientational limit for strongly oriented layered films, with sample geometry, beam phase space, mosaicity, detector resolution, and ordered structure-factor weighting included explicitly. Both materials exhibit the expected 2D-powder signatures: off-specular Bragg arcs from in-plane HK families and specular-family intensity from $HK = 0$ reflections.

For each material, the comparison is shown in two complementary views. The detector-space panel identifies the fixed-Q_r/HK Bragg-rod trajectories used for projection while keeping the measured area-detector intensity visible. The profile panel then gives the quantitative comparison with measured and calculated intensity plotted on the same axes as a function of Q_z for the selected HK families. The detector image is the map of where the profiles are extracted. The plus and minus branches, where shown in profile panels, refer only to the two detector-side intersections of the same in-plane rod family and are not separate crystallographic families.

A central result of the ordered-film comparison is the need for a Lorentzian contribution to the mosaic distribution. The clearest evidence is the $HK = 0$ specular family, i.e. the $00L$ peaks. For a perfectly aligned crystallite at the nominal incidence angle θ_i, only the specular condition directly selected by that incidence geometry should carry appreciable intensity. The measured detector images and projected profiles instead contain additional $HK = 0$ peaks whose Bragg conditions are not directly excited by θ_i. A narrow Gaussian-like mosaic core can account for the strong near-condition arc widths, but it decays too rapidly to supply enough diffracting volume for these off-condition specular peaks. The Lorentzian tail represents the small population of crystallites tilted far enough from the mean film normal to bring those reflections onto the Ewald sphere.

A second important feature appears near the $HK = 0$, $L = 1$ reflection, where the measured image shows a star-like pattern together with bright intensity extending above the critical-angle region. This feature is not assigned to reflectivity. Instead, it is explained within the same diffraction picture by the finite bandwidth of the incident beam, which gives the Ewald sphere an effective thickness. Because the $HK = 0$, $L = 1$ Bragg sphere has comparatively small reciprocal-space radius, this finite-thickness Ewald shell illuminates a relatively large fraction of it. The result is enhanced intensity and the characteristic star-like morphology. This feature therefore provides an additional test that the forward model must treat the experimental phase space, not only the nominal central wavelength and incidence condition.

Figure 5 gathers the ordered-film validation into a single matched comparison. Panels (a) and (b) are the detector-space views for Bi_2Se_3 and Bi_2Te_3, respectively. These panels identify the fixed-Q_r/HK trajectories used for projection and show both off-specular and $HK = 0$ features in the measured images. Panels (c) and (d) show the corresponding projected Q_z profiles for Bi_2Se_3 and Bi_2Te_3. In each profile panel, the measured and calculated intensities have been subjected to the same projection and integration procedure.

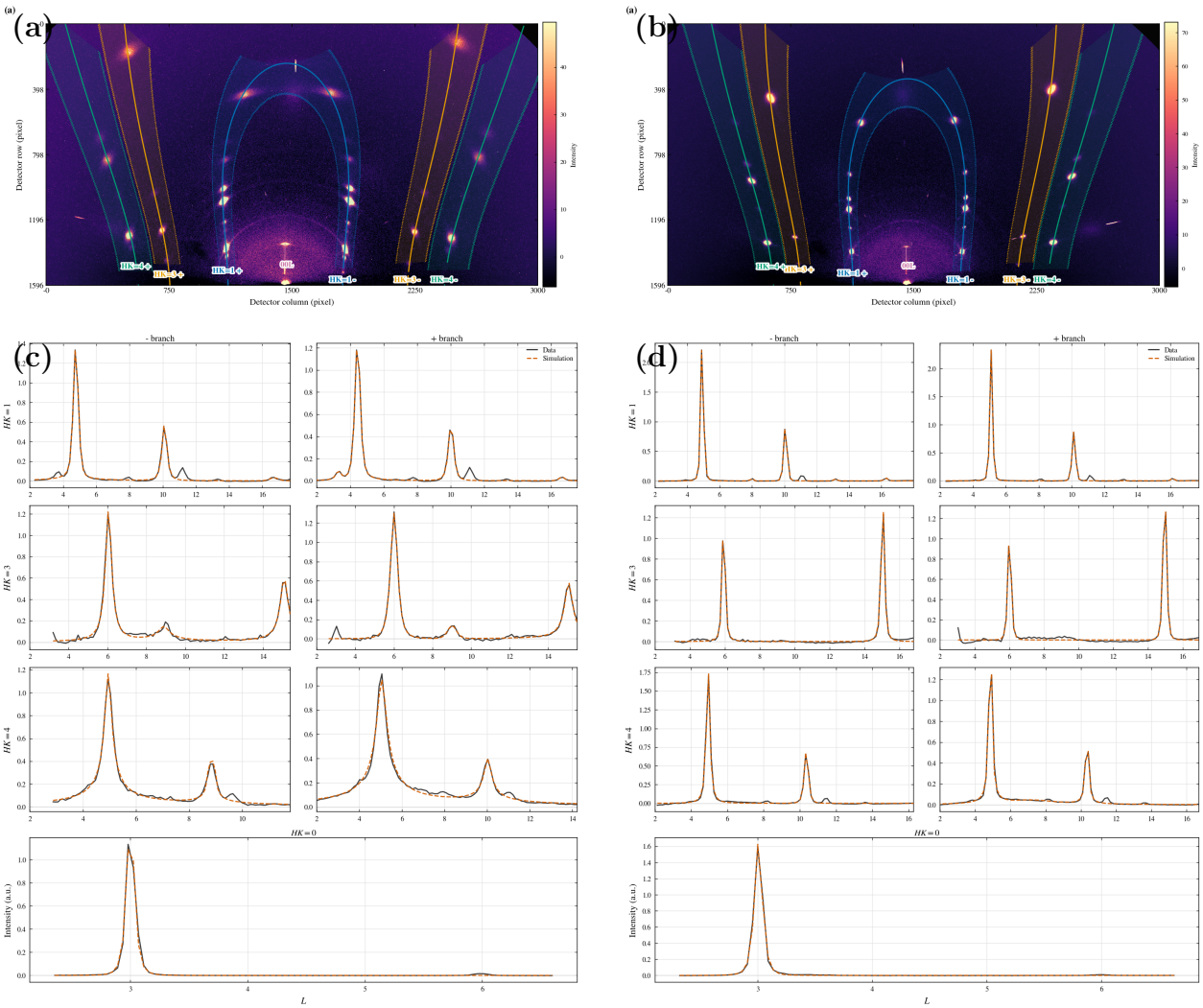

Figure 5: Ordered-film validation for Bi_2Se_3 and Bi_2Te_3. (a,b) Detector-space views show the selected fixed-Q_r/HK trajectories. (c,d) The corresponding Q_z profiles compare measured and calculated intensities using the same projection and integration procedure. The $HK = 0$ rows probe off-condition $00L$ intensity associated with the Lorentzian mosaic tail.

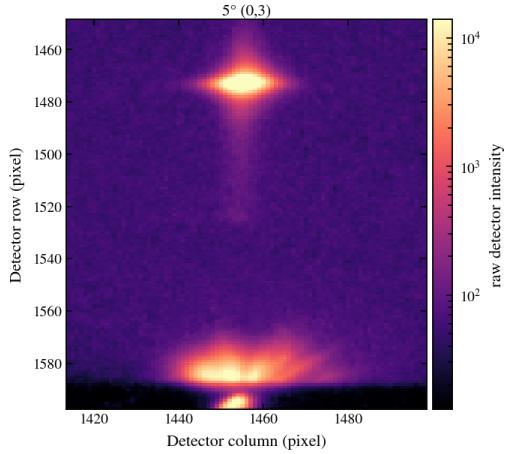

Figure 6: Close-up of the $HK = 0$, $L = 1$ detector feature. The star-like intensity above the critical-angle region is not assigned to reflectivity. Instead, the finite incident-bandwidth gives the Ewald sphere an effective thickness, which illuminates a comparatively large fraction of the small-radius $HK = 0$, $L = 1$ Bragg sphere and produces the enhanced intensity and star-like morphology.

Because mosaicity, beam divergence, wavelength spread, incidence-angle uncertainty, detector resolution, and finite integration width all broaden the effective scattering condition, the projected coordinate should not be interpreted as an ideal reciprocal-space cut from a single perfect crystallite. The comparison remains meaningful because the calculated image is subjected to the same projection and resolution effects before it is compared with the measured profile. The figure therefore tests the detector-space calculation as used experimentally.

In Fig. 5, the off-specular HK profiles and the $HK = 0$ profiles play complementary roles. The off-specular arcs establish that the 2D-powder reciprocal-space construction and calibrated detector geometry put the selected Bragg-rod trajectories in the correct places. Their widths and asymmetries constrain the Gaussian-like mosaic core together with instrumental resolution. The $HK = 0$ peaks that are not directly excited by the nominal θ_i condition then test the Lorentzian tail specifically. A model with only a rapidly decaying mosaic core could fit the main arcs while missing this weak specular-family intensity, so the $HK = 0$ rows provide the visible evidence that the long-tail mosaic population is required.

Figure 6 highlights a second feature that depends on the experimental phase-space treatment. The bright star-like intensity near $HK = 0$, $L = 1$ extends above the critical-angle region, but is not a reflectivity signal. Instead, it arises because the finite bandwidth gives the Ewald sphere an effective thickness, allowing a large portion of the small-radius $HK = 0$, $L = 1$ Bragg sphere to contribute simultaneously. This makes the morphology near $HK = 0$, $L = 1$ an additional check that the forward model must include wavelength bandwidth explicitly.

This combined figure also closes the staged refinement workflow. Detector-space peak positions test the instrumental geometry, incidence condition, sample offsets, and goniome-

ter misalignment. Local widths and tails test the two-component mosaic distribution and resolution terms. The off-condition $HK = 0$ specular peaks test the Lorentzian mosaic contribution. Relative Bragg intensities test the ordered structure-factor description.

5 Discussion and conclusion

The ordered Bi_2Te_3 and Bi_2Se_3 films provide the validation for both the mosaic model and forward modelling. A single detector-space workflow accounts for detector geometry, finite beam phase space, sample alignment, mosaicity, resolution, and ordered structure-factor intensity, then compares measured and calculated profiles along the same indexed Q_r/HK trajectories. The key result is the direct Q_z line-shape agreement.

Keeping the comparison in detector space is essential because the information that constrains the model is distributed between arcs, caps, and weak off-condition specular peaks that would be obscured by a conventional 1D reduction. All these affects are present in the calculated image, so the overlay tests the complete experimental model.

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